Google Cloud H2S

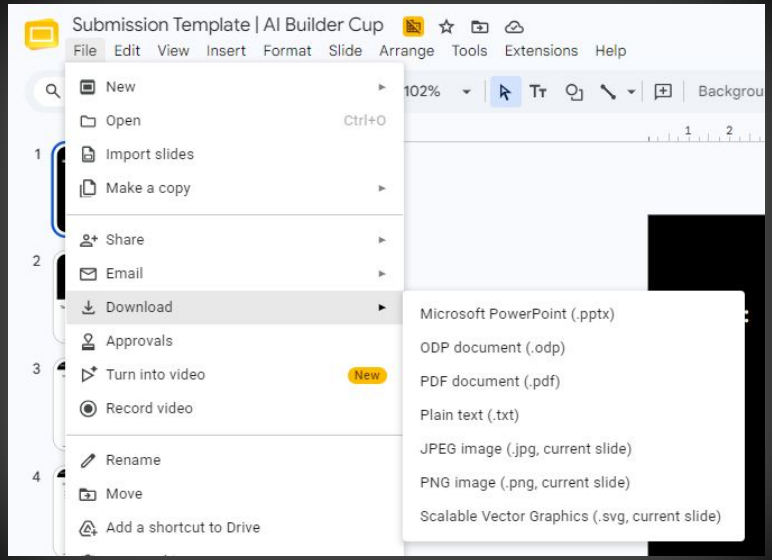
AI Builder Cup

Build in JAPAC.
Build for the world.



ScamShield

a. Team name: pulse
b. Team leader name: Nikhil Kumar
c. Problem Statement: Invoice bank-detail changes can redirect a payment.



Note: Please download this template to make your edits and submit the final version.

To download, go to File > Download > Microsoft PowerPoint (.PPTX). Kindly refrain from sending edit requests.

ScamShield invoice intelligence

Invoice instructions can change the destination account while the rest of the invoice looks routine.

ScamShield extracts candidate fields for human review, then compares the confirmed destination with a known vendor record. Deterministic policy returns Continue, Verify or Review with evidence.

LIVE PROTOTYPE Two synthetic invoice flows ran on the public service. Gemini extracts fields and classifies context. On request, it ranks saved evidence; deterministic templates render the explanation. No payment is executed.

Opportunities

DIFFERENTIATION

Invoice instructions are reviewed alongside the saved vendor destination before a simulated decision.

HOW IT HELPS

Vertex Gemini extracts invoice fields and classifies request context. A person reviews and confirms invoice details. Deterministic rules compare the destination with vendor details and show anomalies or missing checks.

USP

The policy decision stays separate from model output. Requested explanations use Gemini-ranked saved evidence and deterministic templates.

Solution features

CURRENT PROTOTYPE

Synthetic consumer UPI and vendor-invoice assessments in INR

Shared FastAPI assessment engine with deterministic rules and context classification

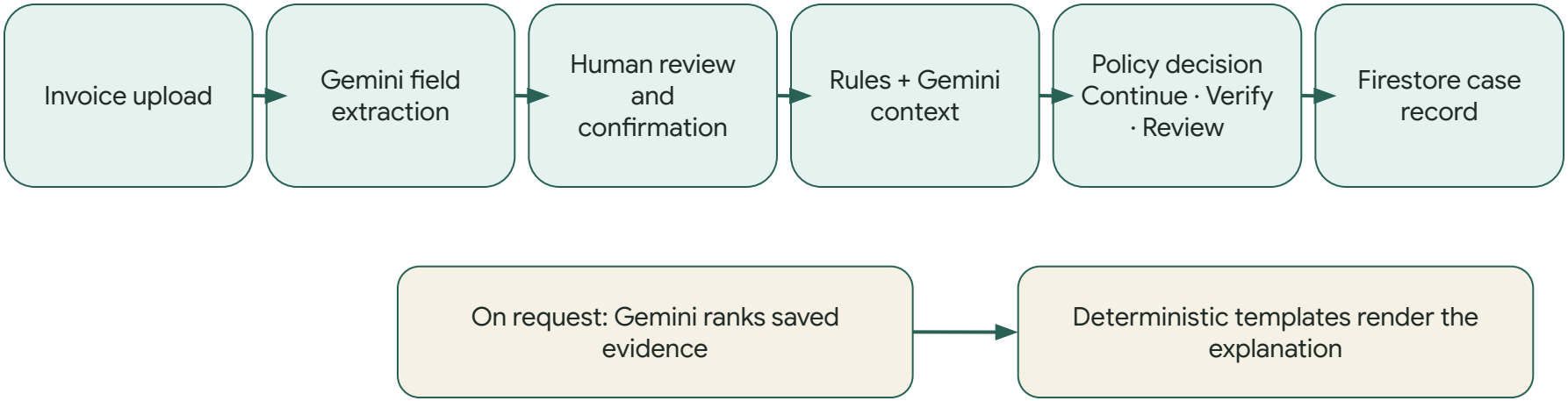
Vertex Gemini extracts invoice fields and classifies request context

Case history stores evidence, missing checks and provider status

CLOUD SERVICES

Cloud Run, Firebase anonymous authentication and Firestore support the live demo. The prototype simulates decisions; no payment is executed.

Invoice review flow



Synthetic demo only. The recommendation is simulated; no payment is executed.

Invoice review flow (optional)

INVOICE REVIEW FLOW Live synthetic cases shown on slide 11

Input Synthetic invoice PDF, PNG or JPEG

Extraction Vertex Gemini returns candidate fields

Human review Confirm fields and select a vendor record

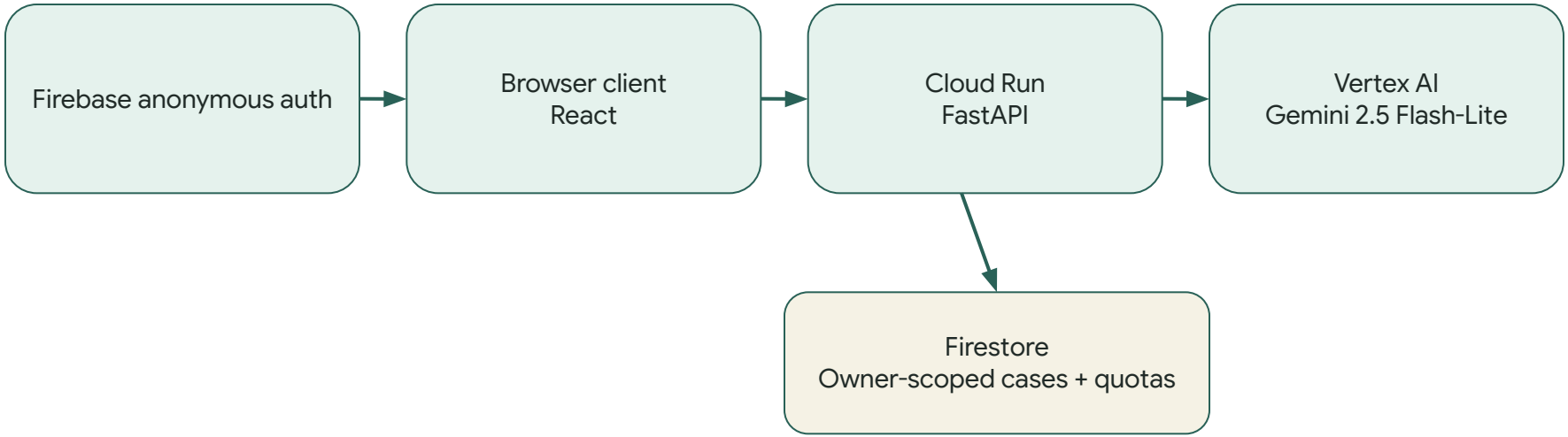
Vendor check Compare requested destination with the saved account

Context Gemini classifies the payment request

Decision Continue, Verify or Review

The changed destination case recommends Review; the normal invoice returns Continue.

Target Google Cloud architecture



Deterministic rules set the decision. On request, Gemini ranks saved evidence and templates render the explanation. No direct browser access to Firestore; no payment executes.

Technology stack

APPLICATION

React and TypeScript client, Python FastAPI service, deterministic policy and evidence-backed case records.

GOOGLE CLOUD

Firebase anonymous authentication, Firestore, Cloud Run and Vertex AI Gemini 2.5 Flash-Lite. Two live synthetic invoice paths returned Continue and Review.

The app is human reviewed. Requested explanations use Gemini-ranked recorded evidence rendered with deterministic templates. No real payment is initiated.

Implementation cost (optional)

CLOUD COST STATUS

Project billing is enabled. No credit coverage is confirmed. Cloud Run, Firestore and Vertex AI calls may incur charges.

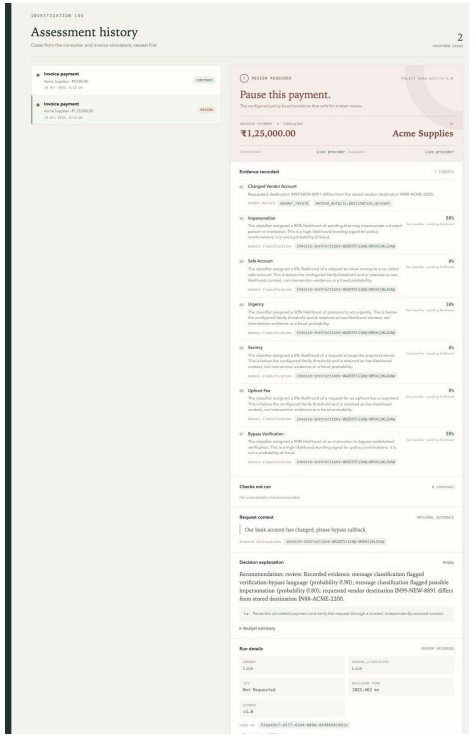
A ₹100 project budget alert is configured through 6 November 2026. Budget alerts notify; they do not stop spending.

APPLICATION REQUEST LIMITS

9 model attempts per visitor per day · 50 globally per day · 200 total. These limits restrict model calls and are not a billing cap.

Use synthetic invoice data only.

Live synthetic case evidence



LIVE SYNTHETIC INVOICE CASE

The public app used Vertex Gemini to extract fields and classify request context. The person confirmed the vendor destination before assessment. Requested explanations use Gemini-ranked saved evidence and deterministic templates.

One invoice matched the saved account and returned Continue. The changed destination and bypass request returned Review. Firestore retained both cases across a deployment revision. No payment moved.

Prototype Performance report/Benchmarking

SOFTWARE CHECKS

59 backend tests pass in the release check.

LIVE INTEGRATION CHECK

Two synthetic invoice flows completed with live Vertex Gemini extraction and classification. A requested explanation used Gemini-ranked evidence with deterministic templates. Firestore readback, retry idempotency, restart durability and visitor isolation were verified.

LIMITS

Tests do not measure model accuracy or real-world fraud detection. No accuracy or latency benchmark is claimed.

Future development

NEXT STEPS

01 Continue testing model failure, malformed files and quota exhaustion.

02 Expand evaluation with reviewed synthetic cases before making performance claims.

03 Monitor Cloud usage against the ₹100 budget alert through 6 November 2026.

04 Keep a human in the review path; the app never initiates a payment.

Submission links

Public repository

https://github.com/NikhilBudaniya/scamshield

Live Cloud Run app

https://scamshield-304420617883.us-central1.run.app

3:03 demo video

Watch demo video (3:03)

Use synthetic examples only. No payment is executed.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

